[scan 25]

B)
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Since $\gamma_2$ vanishes at † $P_\omega$, $\gamma_3$ at the $Q_i$, it follows that there is no invariant section of an $\mathcal{O}(n)^*$ † over $\Sigma_J^*$ † for $n \leq 6$, hence that the order of $\mathrm{cl}(\mathcal{O}(1))$ in $H^1(\Sigma_J, \mathfrak{S}_J; \mathbb{G}_m)$ is indeed 6, which proves that $\mathcal{O}(1)$ is a generator of the mixed $H^1$.

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